

■ A.6.3 Three-Dimensional Parametric Plots

Section 1.8.6 discussed the built-in *Mathematica* function `ParametricPlot`, which generates two-dimensional parametric plots. Here is a package for producing various kinds of three-dimensional parametric plots.

```
ParametricPlot3D::usage =
  "ParametricPlot3D[{x,y,z}, {u,u0,u1,(du)}, {v,v0,v1,(dv)}, (options...)]
  plots a 3D parametric surface. Options are passed to Show[]"

SpaceCurve::usage =
  "SpaceCurve[{x,y,z}, {u,u0,u1,(du)}, (options...)]
  plots a 3D parametric curve. Options are passed to Show[]"

MakePolygons[v1_List] :=
  Block[{l = v1,
    l1 = Map[RotateLeft, v1],
    mesh},
    mesh = {l, l1, RotateLeft[l1], RotateLeft[l]};
    mesh = Map[Drop[#, -1]&, mesh, {1}];
    mesh = Map[Drop[#, -1]&, mesh, {2}];
    Polygon /@ Transpose[ Map[Flatten[#, 1]&, mesh] ]
  ]

plotpoints = PlotPoints /. Options[Plot3D]; (* default *)

Attributes[ParametricPlot3D] = {HoldFirst}

ParametricPlot3D[ fun:{_, _, _},
  ul:{_, u0_, u1_, du_},
  vl:{_, v0_, v1_, dv_},
  opts___] :=
  Show[Graphics3D[MakePolygons[Table[N[fun], ul, vl]]], opts] /;
    NumberQ[N[u0]] && NumberQ[N[u1]] && NumberQ[N[du]] &&
    NumberQ[N[v0]] && NumberQ[N[v1]] && NumberQ[N[dv]]

ParametricPlot3D[fun_, {u_, u0_, u1_}, {v_, v0_, v1_}, opts___] :=
  ParametricPlot3D[ fun,
    {u, u0, u1, (u1-u0)/(plotpoints-1)},
    {v, v0, v1, (v1-v0)/(plotpoints-1)},
    opts ]

Attributes[SpaceCurve] = {HoldFirst}

SpaceCurve[ fun:{_, _, _}, ul:{_, u0_, u1_, du_}, opts___] :=
  Show[ Graphics3D[Line[Table[N[fun], ul]]], opts ] /;
    NumberQ[N[u0]] && NumberQ[N[u1]] && NumberQ[N[du]]
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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```
SpaceCurve[fun_, {u_, u0_, u1_}, opts___] :=
  SpaceCurve[ fun, {u, u0, u1, (u1-u0)/(plotpoints-1)}, opts ]
```

Three-dimensional parametric plots.

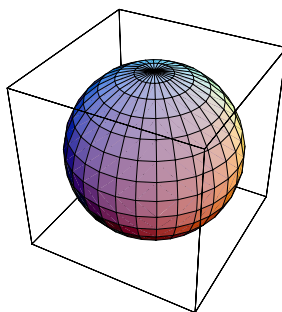
The function `ParametricPlot3D` plots surfaces generated by varying two parameters. The basic way that it works is to make an array of function values, then extract quadruples from the array, and render them as squares.

Read in the package.

```
In[1]:= <<ParametricPlot3D.m
```

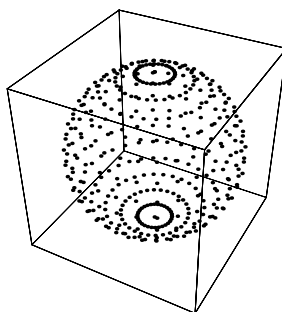
Here is a sphere, generated from a parametric representation. The `BoxRatios` option is passed straight through to `Show`.

```
In[2]:= ParametricPlot3D[
  {Sin[u] Cos[v], Sin[u] Sin[v], Cos[u]},
  {u, 0, Pi, Pi/15}, {v, 0, 2Pi, Pi/15},
  BoxRatios->{1,1,1}]
```



The function `PointParametricPlot3D` is also defined in the package. It draws points on the surface, but does not fill in polygons.

```
In[3]:= PointParametricPlot3D[
  {Sin[u] Cos[v], Sin[u] Sin[v], Cos[u]},
  {u, 0, Pi, Pi/15}, {v, 0, 2Pi, Pi/15},
  BoxRatios->{1,1,1}]
```



Here is an example of a curve in space, specified by a single parameter.

```
In[4]:= SpaceCurve[{u Sin[u], u Cos[u], u},  
                  {u, 0, 15, 0.15}, BoxRatios->{1,1,1}]
```

